1)

30a)

① ∀ object x, if x is above all triangles, then x is above all the blue objects

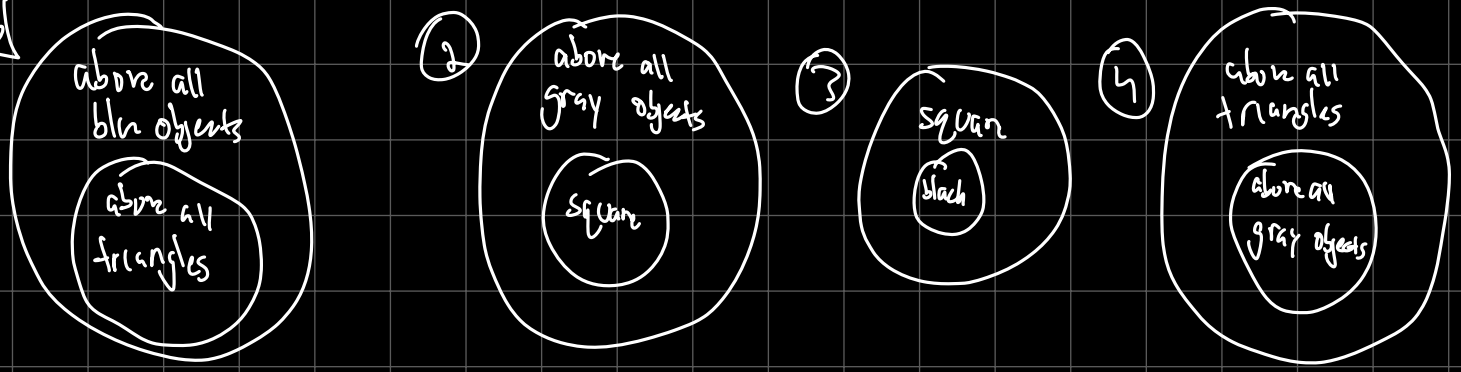
② ∀ object x, if x is a square, then x is above all the gray objects (Contrapos.)

③ ∀ object x, if x is black, then x is a square

④ ∀ object x, if x is above all the gray objects, then x is above all the triangles

∴ ∀ object x, if x is black, then x is above all the blue objects.

30b)



30c) ③ ∀ object x, if x is black, then x is a square

② ∀ object x if x is a square, then x is above all the gray objects (Contrapos)

④ ∀ object x, if x is above all the gray objects, then x is above all the triangles

① ∀ object x if x is [illegible] k then x is above all the blue obje

∴ ∀ object x, if x is black, then x is above all the blue objects.

2)

#32a)

① If I do not understand an example, then I grumble at the example (contrapos.)

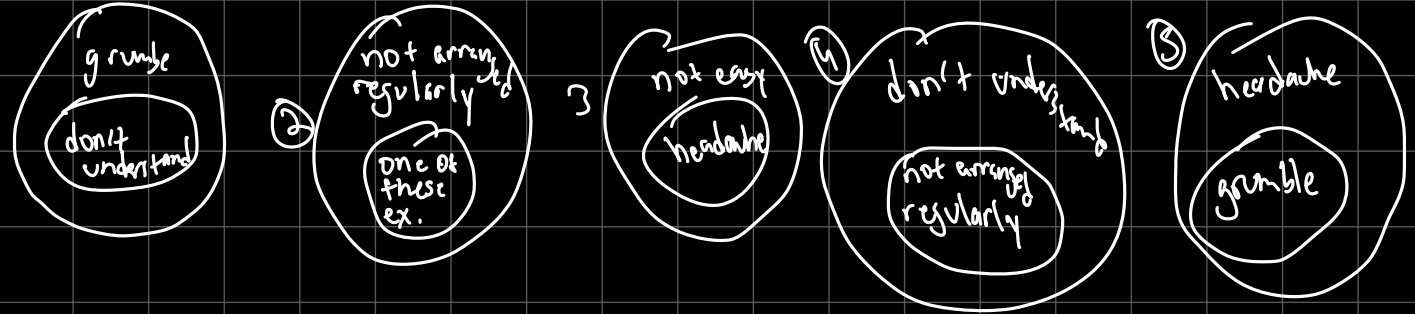
② If the argument is one of these examples, then its arguments are not arranged in regular order

③ If an example gives me a headache, then it is not easy (Contrapos.)

④ If an arguments arguments are not arranged in regular order, then I don't understand ex

⑤ If I grumble at an example, then it gives me a headache.

3#b)



3#c)

② If the argument is one of these examples, then its arguments are not arranged in regular order.

④ If an argument's arguments are not arranged in regular order, then I don't understand ex.

① If I do not understand an example, then I grumble at the example (contrapos.)

⑤ If I grumble at an example, then it gives me a headache.

③ If an example gives me a headache, then it is not easy (contrapos.)

∴ If the argument is one of these examples, then it is not easy.

#3a)

① ∀ writer x, if x understands human nature, then x is clever

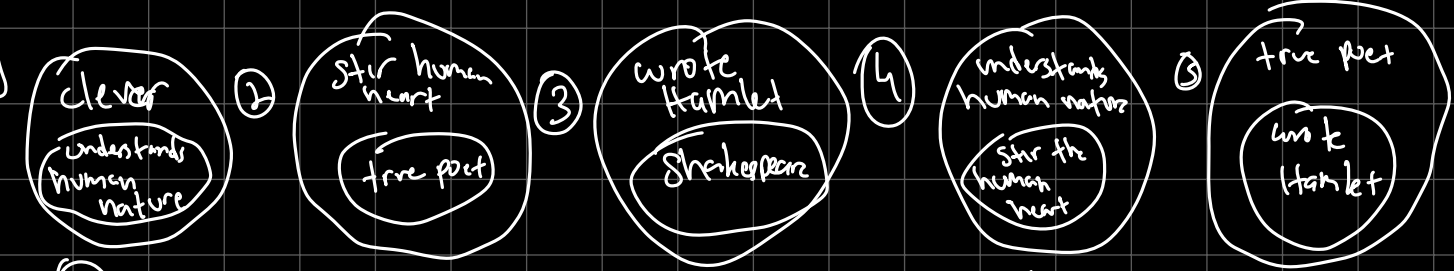
② ∀ person x, if x is a true poet, then x can stir the human heart

③ If x is Shakespeare, then x wrote Hamlet

④ ∀ writer x, if x can stir the human heart, then x understands human nature (Contra pos.)

⑤ ∀ person x, if x wrote Hamlet, then x is a true poet

3#b)



3#c) ③ If x is Shakespeare, then x wrote Hamlet

⑤ ∀ person x, if x wrote Hamlet, then x is a true poet

② ∀ person x, if x is a true poet, then x can stir the human heart

④ ∀ writer x, if x can stir the human heart, then x understands human nature (Contra pos.)

① ∀ writer x, if x understands human nature, then x is clever

∴ Shakespeare is clever

3) 7.11, 16

#7) Let $a=0$ and $b=1$, then:

$$\sqrt{a+b} = \sqrt{0+1} = \sqrt{1} = \pm 1$$ choosing '+1', so, +1

$$\sqrt{a}+\sqrt{b} = \sqrt{0}+\sqrt{1} = 0 \pm 1$$ so, +1

$$+1 = +1$$

True

#11) let $n=1$, then $2n^2-5n+2 = 2(1)^2-5(1)+2$

$$= 2-5+2 = -1$$

-1 is a prime integer

#16) let $n=8$, then $n^2+1 = 8^2+1 = 64+1 = 65$

$65 = 1 \times 65 = 5 \times 13$ so 65 is not prime

Therefore, the statement is false because $n=8$ is even, but 8^2+1 is not prime

4] 22, 29

#22] $n^2 - n + 11$

$n=1$: $1 - 1 + 11 = 11$ ✓
$n=2$: $4 - 2 + 11 = 13$ ✓
$n=3$: $9 - 3 + 11 = 17$ ✓
$n=4$: $16 - 4 + 11 = 23$ ✓
$n=5$: $25 - 5 + 11 = 31$ ✓
$n=6$: $36 - 6 + 11 = 41$ ✓
$n=7$: $49 - 7 + 11 = 53$ ✓
$n=8$: $64 - 8 + 11 = 67$ ✓
$n=9$: $81 - 9 + 11 = 83$ ✓
$n=10$: $100 - 10 + 11 = 101$ ✓

Since these are all prime, statement is true

#29] a. def. of even integer
b. Substitution
c. integers are closed under addition
d. def. of even integer

5] 8, 9, 14

#8] Suppose m and n are any integers such that m is even and n is odd. By definition of odd, this means we must show $5m + 3n = 2(\text{some integer}) + 1$

By def of even, $m = 2r$, for some integer r.
By def of odd, $n = 2s + 1$, for some integer s.
Then $5m + 3n = 5(2r) + 3(2s+1)$ (by substitution)
$= 10r + 6s + 3$
$= 10r + 6s + 2 + 1$

$= 2(5r+3r+1) + 1$ (by algebra)

let $k = 5r+3s+1$

Since r and s are integers, k is an integer b/c products and sums of integers are integers.

Thus $5m+3n = 2k+1$, when k is an integer

Then $5m+3n$ = odd by def of odd

9| Suppose n is an integer greater than 4 and n is a perfect square.

Then, $n = k^2$ for some integer k.

Hence, $n-1 = k^2-1$ (by algebra)

$= (k+1)(k-1)$ (by algebra)

Since $n > 4$, $k^2 > 4$ (substitution)

So, $|k| > 2$. Therefore $n-1 = (k-1)(k+1)$ is a product of two integers, indicating $n-1$ is not prime.

14| To prove the given statement is false, we prove that its negation is true.

Negation is: "For every integer $k \geq 4$, $2k^2-5k+2$ is not prime."

Proof of negation: Suppose k is any integer with $k \geq 4$.

By algebra, $2k^2-5k+2 = (2k-1)(k-2)$.

Because $k \geq 4$, $2k-1 \geq 7$ and $k-2 \geq 2$

Thus $2k-1$ and $k-2$ are both pos integers greater than 1

So $2k^2-5k+2$ is not prime and original statement is false.

6)

#18) The proof assumes what is to be proved. It says "If mn is even, then $mn = 2r$". The purpose is to prove that mn is even, but that hasn't been proved yet. So the argument is circular and the proof is invalid

#19) The mistake in the proof is that the same symbol k is used to represent 2 possibly different integers. From m being even, we may write $m = 2p$ for some integer p. For some integer p, and from n being even, $n = 2q$ for some integer q

But writing $m = 2k$ and $n = 2k$ implies that $m = n$. So the proof only considers cases where m and n are the same value, rather than 2 arbitrarily chosen even integers. So the proof is invalid.

7) 26, 30, 31

#26) False. Counter-example; let $a = 2, b = 3, c = 4$. These are consecutive integers, but $a+b+c = 2+3+4 = 9$, and 9 is odd. So the statement is false

#30) False. CE: Let $m = 3$. Then $m > 2$, but $m^2 - 4 = 3^2 - 4 = 9 - 4 = 5$. Since 5 is prime, $m^2 - 4$ is not composite and statement is false

31) False. CE. Let $n = 11$. Then $n^2 - n + 11 = 11^2 - 11 + 11 = 121$. But $121 = 11 \cdot 11$. So 121 is not prime and the statement is False